

KENDRIYA VIDYALAYA SANGATHAN BHUBANESWAR REGION
FIRST PRE-BOARD EXAMINATION NOV.-2024
CLASS X (SESSION 2024-2025)
MARKING SCHEME SET-1

ARTIFICIAL INTELLIGENCE (SUBJECT CODE - 417)

Time Allowed : 2 Hours

Maximum Marks : 50

SECTION A: OBJECTIVE TYPE QUESTIONS

Question	Answer	Marks
Q1		
i	(iii) Fantastic video! You did an excellent job. You chose accurate images with extremely calming background music. Your explanation of the topic with perfect pronunciation was also quite effective.	1
ii	(b) Financial difficulties	1
iii	(d) Chatbot	1
iv	(c) Double clicking	1
v	(c) It can overheat	1
vi	(c) By buying jute from the local farmer and by providing jobs to local women	1
Q.2		
i	(a) Google Maps, Ola, Uber	1
ii	Naturalist Intelligence	1
iii	(b) (ii) and (iii)	1
iv	(a) Training Data	1
v	(a) Data Privacy	1
vi	(c) Pixel	1
Q.3		
i	Problem Statement Template	1
ii	(b) Hidden layer	1
iii	(d) Neural networks	1
iv	(i) (a) and (d) only	1
v	(a) 17	1
vi	(d) All of the mentioned	1
Q.4		
i	Text Classification	1
ii	(d) Text and Speech	1
iii	Corpus	1
iv	Computer Vision	1
v	(iii) Clustering -> unknown data set	1
vi	Script bot	1
Q.5		
i	Token	1
ii	Confusion Matrix	1
iii	(a) False Positive	1
iv	(b) Precision and Recall	1

v	(iii) Smart Chatbot	1
vi	(a) Accuracy	1
SECTION B: SUBJECTIVE TYPE QUESTIONS Each Answer question in 20 – 30 words.		
Q.6	In SMART goals, A refers for Achievable, it means breaking down big goals into smaller parts will make the goal achievable. For example Bigger Goal: "I want to become a teacher in my school." Breaking it into smaller goals: Complete higher secondary Complete Graduation Complete B.Ed. Apply for jobs in the teaching field <i>(1 mark for identification; 1 mark for explanation)</i>	2
Q.7	The four steps of effective time management which Ekta must have followed are: (i) Organise (ii) Prioritise (iii) Control (iv) Track <i>(½ mark for every step, ½ * 4 =2)</i>	2
Q.8	Two methods how virus get into the computer: 1. Using pen drive without scanning. 2. Not installing Anti-virus and Firewall 4. No opening Secure sites rather opening unsecure sites <i>(1 mark for each correct method, 1*2=2)</i>	2
Q.9	I. Fulfil Customer Needs II. Use Local Materials III. Help Society IV. Create Jobs V. Sharing of Wealth VI. Lower Price of Products <i>(1 mark for each correct point, 1*2=2)</i>	2
Q.10	This stage is the Survive stage of an entrepreneur's career process. In this stage, even if there are many entrepreneurs in the market, the new entrepreneur has to remain in a competitive market. <i>(1 mark for mentioning the stage; 1 mark for correct explanation)</i>	2
Answers in 20 – 30 words each (2 x 4 = 8 marks)		
Q.11	Any machine that has been trained with data and can make decisions/predictions on its own can be termed as AI. Eg: The bot or the automation machine is not trained with any data is not an AI while a chatbot that understands and processes human language is an AI. <i>(1 mark for correct explanation; ½ mark for example of AI; ½ mark for example of not AI)</i>	2
Q.12	In the given scenario, we are concerned about the bias. When we talk about a machine, we know that it is artificial and cannot think on its own. It can have intelligence, but we cannot expect a machine to have any biases of its own. Any bias can transfer from the developer to the machine while the algorithm is being developed. <i>(1 mark for mentioning the term bias; 1 mark for the correct explanation)</i>	2
Q.13	Steps of AI project life cycle: 1. Data Acquisition 2. Data Exploration 3. Modelling 4. Evaluation	2

	<i>(½ mark for mentioning each stage, ½ *4=2)</i>	
Q.14	Stemming Lemmatization happily happi happy Process that takes longer time for execution is lemmatization <i>(½ marks each for identifying the correct stem and lemma; 1 mark for identifying the correct process)</i>	2
Q.15	Bag of words gives us two things: 1. A vocabulary of words for the corpus 2. The frequency of these words (number of times it has occurred in the whole corpus) <i>(1 mark for each point)</i>	2
Q.16	Let us take each of the factor into consideration at once, If precision is considered, FN cases will not be taken into account, so it will be of great loss as if the machine will predict there will be no heavy rain, but if the rain occurred, it will be a big monetary loss due to damage to crops. If only recall is considered, then FP cases will not be taken into account. This situation will also cause a big amount of loss, as all people of the village are dependent on farmers for food, and if the model predicts there will be heavy rain and the farmers may not grow crops, it will affect the basic needs of the people. Hence F1 Score is the best suited parameter to test this AI model, which is the balance between Precision and Recall. <i>(1 mark for identifying the term F1 score; 1 mark for relevant explanation)</i>	2
Answers in 50– 80 words each (4 x 3 = 12 marks)		
Q.17	All humans possess 9 types of intelligence but at different levels. They are: 1. Mathematical Logical Reasoning: ability to regulate, measure, and understand numerical symbols, abstraction and logic. 2. Linguistic Intelligence: Language processing skills both in terms of understanding or implementation in writing or verbally. 3. Spatial Visual Intelligence : ability to perceive the visual world and the relationship of one object to another. 4. Kinaesthetic Intelligence : ability that is related to how a person uses his limbs in a skilled manner. 5. Musical Intelligence : ability to recognize and create sounds, rhythms, and sound patterns. 6. Intrapersonal Intelligence : Describes how high the level of self-awareness someone has is. Starting from realizing weakness, strength, to his own feelings. 7. Existential Intelligence : An additional category of intelligence relating to religious and spiritual awareness. 8. Naturalist Intelligence : An additional category of intelligence relating to the ability to process information on the environment around us. 9. Interpersonal intelligence : ability to communicate with others by understanding other people's feelings & influence of the person. <i>(½ mark for the naming the intelligence; ½ mark for the explanation of the same; (½ + ½) * 4 = 4)</i>	4
Q18	Artificial Intelligence (AI) refers to any technique that enables computers to mimic human intelligence i.e., make decisions, predict the future, learn and improve on its own. With respect to the type of data fed in the AI model, AI models can be broadly categorised into three domains: 1. Data sciences 2. Computer vision 3. Natural Language Processing Data Science takes input in the form of numeric and alphanumeric data. Computer Vision takes input in the form of images and videos. Natural Language Processing takes input in the form of text and speech. <i>(1 mark for definition of AI; ½ mark each for the names of the domains; ½ mark each for the type of data input to domains)</i>	4

Q.19	Neural networks are loosely modelled after how neurons in the human brain behave. The features of a neural network are : 1. They are able to extract data features automatically without needing the input of the programmer. 2. A neural network is essentially a system of organizing machine learning algorithms to perform certain tasks. 3. It is a fast and efficient way to solve problems for which the dataset is very large, such as in images. <i>(1 mark for how neural networks are modelled; 1 mark each for relevant feature of neural network)</i>	4
Q.20	1. Tokenisation Akash, and, Ajay, are, best, friends Akash, likes, to, play, football, but, Ajay, prefers, to, play, online, games 2. Removal of stopwords Akash, Ajay, best, friends Akash, likes, play, football, Ajay, prefers, play, online, games 3. converting text to a common case akash, ajay, best, friends akash, likes, play, football, ajay, prefers, play, online, games 4. Stemming/Lemmatisation akash, ajay, best, friend akash, like, play, football, ajay, prefer, play, online, game <i>(1 mark for each step; 1*4=4)</i>	4
Q.21	A chatbot is a software application designed to simulate human conversation, allowing users to interact through text or voice. It uses Natural Language Processing (NLP) and Machine Learning (ML) technologies to understand, process, and respond to user queries in a conversational manner. Some chatbots are rule-based (responding to specific triggers), while others use artificial intelligence to provide more nuanced and flexible responses. Technology behind a chatbot: 1. Natural Language Processing (NLP) – helps chatbots understand and generate human language. 2. Machine Learning (ML) – enables chatbots to learn from data, improving responses over time. Examples of Chatbots: 1. ChatGPT by OpenAI 2. Siri by Apple Or any two chatbots 2 marks for correct definition and 2 marks for correct examples	4
